

Labwork 2

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I. INTRODUCTION

This report details the implementation of a U-Net based segmentation model for measuring fetal head circumference from ultrasound images as part of the second labwork.

II. DATA DESCRIPTION

The dataset is sourced from the HC18 Grand Challenge. It contains 1,334 two-dimensional ultrasound images of the fetal head standard plane. The collection consists of 999 training images and 335 test images.

Each image has the resolution of 800×540 pixels with pixel size ranging from 0.052 to 0.326 mm. Pixel sizes are provided via CSV files. For the training set, manual annotations and head circumference measurements (in mm) are included.

In this report, the original test set is not used due to the absence of public labels. Instead, the 999 training images are split into 899 images for training and 100 images for validation and performance measurement.

III. MODEL ARCHITECTURE

The model utilizes the U-Net architecture from the `segmentation-models.pytorchlibrary` [?]. A ResNet50 backbone with ImageNet pretrained weights serves as the encoder. Since the input images are grayscale, the model is configured with a single input channel. For binary segmentation, the output class is set to 1. Spatial and channel squeeze-and-excitation (scSE) attention modules are integrated into the decoder.

IV. TRAINING

Images are resized to 512×512 pixels. Data augmentation includes random horizontal flips ($p = 0.5$), shifting, scaling, and rotation. Pixel values are normalized and converted to tensors using `Albumentations` and `torchvision`.

The model is training for 40 epochs with a batch size of 8. We utilize the DiceLoss function and the AdamW optimizer with a learning rate of $1e - 4$ and weight decay of $1e - 5$. Mixed precision training is implemented via `GradScaler`.

V. RESULT

Post-processing involves `cv2.findContours` to extract the mask boundary. These contours are used to fit an ellipse via `cv2.fitEllipse`. The final head circumference is calculated using Ramanujan's approximation for ellipse circumference.

On the 100-image validation set, the model achieved an absolute mean difference and standard deviation of 23.76 ± 12.67 mm. This performance is lower than the top tier results on the

HC18 leaderboard, indicating a need for further hyperparameter tuning or architectural refinement.

REFERENCES

- [1] P. Yakubovskiy, "Segmentation Models Pytorch," GitHub, 2020. [Online]. Available: https://github.com/qubvel/segmentation_models.pytorch.
- [2] "HC18 - Grand Challenge," [Online]. Available: <https://hc18.grand-challenge.org/>.